

Temperature control

On the shores of Antarctica, winter temperatures can drop to -40°F (-40°C), and storm-force winds make it feel colder still. Despite this, penguins keep their bodies at 100°F (38°C)—the same temperature as birds in the jungle. Like all birds and mammals, they are warm-blooded, or endothermic. Their muscles and livers keep them warm by acting like chemical furnaces, fueled by energy from food. Most other animals are cold-blooded, or ectothermic. Instead of generating their warmth internally, they get it from outside. In some habitats, such as the deep sea, cold-blooded animals are always cold. But in warm places, they sometimes get too hot and have to seek shelter in the shade.

Soaking up sunshine

Lizards move slowly when they are cold, so they need a kick-start of warmth before they can start to hunt. By basking in the morning sunshine, a lizard can raise its body temperature to 86°F (30°C), even when the air is still cool. As the day heats up, the lizard keeps its temperature steady by moving in and out of the shade. At sunset, it crawls into a safe crevice, where its body cools down for the night.



Ocellated lizard



Well padded

Stretched out on a beach, a male elephant seal seems to slump over the sand. Its oversized outline is produced by a thick jacket of body fat, or blubber, which keeps it warm when it dives. Blubber is poor at conducting heat, so it stops the seal's warmth from ebbing away into the cold water outside. On land, this layer of fat can easily make the seal overheat. To prevent this from happening, the seal's body automatically diverts blood into its skin, letting surplus heat escape into the air.



Cooling off

In warm weather, kangaroos lounge under trees or rocks, where they are shaded from the sun. But if the air is hot, they use an emergency cooling system. They lick their forearms until the fur is wet. The saliva then evaporates, and the evaporation sucks heat from blood flowing through the arms. Many other mammals use evaporation to keep cool. Dogs and foxes cool down by panting, while horses and large antelopes sweat, just like humans do. Many birds pant by fluttering their throats, but some have a stranger way of losing heat: they squirt liquid droppings onto their legs.

